

MAM4000W Analysis II – Theorems, Lemmas, and Corollaries

Contents

1	Measure Theory and Integration	1
2	Normed Spaces and Linear Mappings	3
3	Hahn–Banach Theorem and Its Consequences	5
4	Baire Category and Banach’s Theorems	6
5	Projection on Convex Sets and Variational Inequalities	7

1 Measure Theory and Integration

Measure Theory

Theorem (Theorem 2.6). *Let $(X, \mathcal{E})$ be a measurable space and $\mu : \mathcal{E} \rightarrow [0, \infty]$ an additive function different from the constant function ∞ . Then:*

[(i)]

1. μ is a positive measure if and only if for any increasing sequence $(E_n)_{n \in \mathbb{N}} \subset \mathcal{E}$,

$$\mu\left(\bigcup_{n=0}^{\infty} E_n\right) = \lim_{n \rightarrow \infty} \mu(E_n) = \sup_n \mu(E_n).$$

2. If μ is a positive measure, then for any decreasing sequence $(E_n)_{n \in \mathbb{N}} \subset \mathcal{E}$ with $\mu(E_0) < \infty$,

$$\mu\left(\bigcap_{n=0}^{\infty} E_n\right) = \lim_{n \rightarrow \infty} \mu(E_n) = \inf_n \mu(E_n).$$

3. If for every decreasing sequence $(E_n)_{n \in \mathbb{N}} \subset \mathcal{E}$ satisfying $\mu(\bigcap_n E_n) = 0$ we have $\lim_{n \rightarrow \infty} \mu(E_n) = 0$, then μ is a positive measure.

Theorem (Theorem 2.14). *Let $(X, \mathcal{E})$ be a measurable space and let $(f_h)_{h \in \mathbb{N}}$ be a sequence of measurable functions $f_h : X \rightarrow \mathbb{R}$. Then the functions $\sup_h f_h$, $\inf_h f_h$, $\limsup_{h \rightarrow \infty} f_h$, and $\liminf_{h \rightarrow \infty} f_h$ are all measurable.*

Corollary (Corollary 2.15). *Let $(X, \mathcal{E})$ be a measurable space and let $(f_h)_{h \in \mathbb{N}} : X \rightarrow \mathbb{R}$ be a sequence of measurable functions. If $f_h(x) \rightarrow f(x)$ pointwise for every $x \in X$ as $h \rightarrow \infty$, then the limit function $f : X \rightarrow \mathbb{R}$ is measurable.*

Theorem (Theorem 2.16). *Let $(X, \mathcal{E})$ be a measurable space and let $f, g : X \rightarrow \mathbb{R}$ be measurable functions. Then:*

1. the functions $\max\{f, g\}$, $\min\{f, g\}$, $f + g$, and fg are measurable;
2. the positive and negative parts $f^+ := \max\{f, 0\}$ and $f^- := \max\{-f, 0\}$ are measurable;
3. the absolute value $|f|$ is measurable.

Theorem (Theorem 2.17). *Let $(X, \mathcal{E})$ be a measurable space. A function $f : X \rightarrow [0, \infty)$ is measurable if and only if it is the pointwise limit of an increasing sequence of simple functions $(\varphi_n)_{n \in \mathbb{N}}$ on X . In other words, there exists a sequence of simple functions $\varphi_n : X \rightarrow [0, \infty)$ such that $\varphi_n(x) \leq \varphi_{n+1}(x)$ for all x and $\varphi_n(x) \rightarrow f(x)$ as $n \rightarrow \infty$ for each $x \in X$.*

Integration

Theorem (Theorem 3.2). *Let $(X, \mathcal{E}, \nu)$ be a measure space. If $f, g : X \rightarrow [0, \infty)$ are two positive simple functions and $t \geq 0$, then:*

[(1)]

$$1. \int_X (f + g) d\nu = \int_X f d\nu + \int_X g d\nu.$$

$$2. \int_X (t f) d\nu = t \int_X f d\nu.$$

$$3. \text{ If } f \leq g, \text{ then } \int_X f d\nu \leq \int_X g d\nu.$$

Theorem (Theorem 3.5 (Beppo–Levi Monotone Convergence)). *Let $(X, \mathcal{E}, \mu)$ be a measure space and let $f_n : X \rightarrow [0, \infty)$ be an increasing sequence of measurable functions (i.e. $f_n(x) \leq f_{n+1}(x)$ for all n and x). If $f(x) := \lim_{n \rightarrow \infty} f_n(x)$ for each $x \in X$, then*

$$\lim_{n \rightarrow \infty} \int_X f_n d\mu = \int_X f d\mu.$$

Theorem (Theorem 3.8 (Fatou's Lemma)). *Let $(X, \mathcal{E}, \mu)$ be a measure space and let $f_n : X \rightarrow [0, \infty)$ be a sequence of measurable functions. Then*

$$\int_X \left(\liminf_{n \rightarrow \infty} f_n(x) \right) d\mu(x) \leq \liminf_{n \rightarrow \infty} \int_X f_n d\mu.$$

Theorem (Theorem 3.12). *Let $(X, \mathcal{E}, \mu)$ be a measure space.*

1. *If $f : X \rightarrow \mathbb{R}$ is μ -integrable and $g : X \rightarrow \mathbb{R}$ is μ -summable (i.e. $|g|$ is integrable), then $f + g$ is μ -integrable and $\int_X (f + g) = \int_X f + \int_X g$.*
2. *If f is μ -integrable and $\alpha \in \mathbb{R}$, then αf is μ -integrable and $\int_X \alpha f = \alpha \int_X f$.*
3. *If f and g are μ -integrable and $f \leq g$ μ -a.e., then $\int_X f \leq \int_X g$.*

Theorem (Theorem 3.14 (Dominated Convergence Theorem)). *Let $(X, \mathcal{E}, \mu)$ be a measure space and let $f_n : X \rightarrow \mathbb{R}$ be a sequence of μ -summable functions converging pointwise to a function $f : X \rightarrow \mathbb{R}$. If there exists an integrable function $g \in L^1(X)$ such that $|f_n(x)| \leq g(x)$ for all n and all $x \in X$, then f is μ -summable and*

$$\lim_{n \rightarrow \infty} \int_X f_n d\mu = \int_X f d\mu.$$

Theorem (Theorem 3.15). *Let $(X, \mathcal{E}, \mu)$ be a measure space and let $f, g : X \rightarrow \mathbb{R}$ be two functions such that $f(x) = g(x)$ for μ -almost every $x \in X$. Then:*

1. *f is μ -integrable (resp. μ -summable) if and only if g is μ -integrable (resp. μ -summable).*
2. *In the case that both are integrable, $\int_X f d\mu = \int_X g d\mu$.*

Theorem (Theorem 3.16). *Let $(X, \mathcal{E}, \mu)$ be a measure space and let (Y, d) be a metric space. If $f : X \times Y \rightarrow \mathbb{R}$ satisfies:*

- *for each fixed $y \in Y$, the function $x \mapsto f(x, y)$ is measurable and integrable on X ;*
- *for each fixed $x \in X$, the function $y \mapsto f(x, y)$ is continuous on Y ;*
- *there exists an integrable function $h(x)$ on X such that $|f(x, y)| \leq h(x)$ for all $y \in Y$ and all $x \in X$;*

then the function $F : Y \rightarrow \mathbb{R}$ defined by $F(y) := \int_X f(x, y) d\mu(x)$ is continuous on Y .

Theorem (Theorem 3.17 (Differentiation Under the Integral Sign)). *Let $(X, \mathcal{E}, \mu)$ be a measure space and (Y, d) a metric space. Suppose $f : X \times Y \rightarrow \mathbb{R}$ satisfies:*

- *for each $t \in Y$, $x \mapsto f(x, t)$ is integrable on X , and for each $x \in X$, the map $t \mapsto f(x, t)$ is differentiable on Y ;*
- *the partial derivative $\partial f / \partial t(x, t)$ exists for all x, t and there is an integrable function $H(x)$ such that $|\frac{\partial f}{\partial t}(x, t)| \leq H(x)$ for all $t \in Y$;*

then $F(t) := \int_X f(x, t) d\mu(x)$ is differentiable for $t \in Y$ and

$$F'(t) = \frac{d}{dt} \int_X f(x, t) d\mu(x) = \int_X \frac{\partial f}{\partial t}(x, t) d\mu(x).$$

Theorem (Theorem 3.21). *Under the above notations, for any $f \in L^\infty(X)$ the quantity*

$$\|f\|_\infty := \mu\text{-ess-sup}_{x \in X} |f(x)|$$

equals the essential supremum of $|f|$ and defines the usual L^∞ norm of f .

Theorem (Theorem 3.23). *For a summable function $f \in L^1(X)$, the quantity*

$$\|f\|_1 := \int_X |f(x)| d\mu(x)$$

defines a norm on the space $L^1(X)$ of integrable functions.

Theorem (Theorem 3.26 (Hölder's Inequality)). *Let $p \in [1, \infty]$ and q be the conjugate exponent (i.e. $\frac{1}{p} + \frac{1}{q} = 1$). If $f \in L^p(X, \mu)$ and $g \in L^q(X, \mu)$, then $fg \in L^1(X, \mu)$ and*

$$\int_X |fg| d\mu \leq \|f\|_p \|g\|_q.$$

Theorem (Theorem 3.27 (Minkowski's Inequality)). *If $p \in [1, \infty]$ and $f, g \in L^p(X, \mu)$, then*

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p.$$

2 Normed Spaces and Linear Mappings

Finite-Dimensional Spaces and Compactness

Theorem (Theorem 3.4). *Let $(V, \|\cdot\|)$ be a normed space of finite dimension. Then a subset $K \subset V$ is compact if and only if K is closed and bounded.*

Lemma (Lemma 3.5 (Riesz's Lemma)). *Let $(V, \|\cdot\|)$ be a normed space over $K = \mathbb{R}$ or $\mathbb{C}$. Let Y be a closed subspace of V , and let Z be another subspace of V with $Y \subsetneq Z$. Then for every $0 < \theta < 1$, there exists a vector $z_\theta \in Z$ such that*

$$\|z_\theta\| = 1, \quad \text{and} \quad \|z_\theta - y\| \geq \theta \quad \forall y \in Y.$$

Theorem (Theorem 3.6). *If a normed space $(V, \|\cdot\|)$ has the property that the closed unit ball $B(0, 1) \subset V$ is compact, then V is finite-dimensional.*

Banach Spaces and Continuous Linear Maps

Theorem (Theorem 4.1). *A normed space $(V, \|\cdot\|)$ is a Banach space (complete) if and only if it satisfies the following series convergence condition: for every sequence $(v_h)_{h \in \mathbb{N}} \subset V$ such that $\sum_{h=0}^{\infty} \|v_h\| < \infty$, the series $\sum_{h=0}^{\infty} v_h$ converges in V .*

Theorem (Theorem 5.3). *Let $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ be normed spaces, and let $L : V \rightarrow W$ be a linear mapping. The following statements are equivalent:*

1. *L is continuous (as a function between metric spaces).*

2. L is continuous at 0_V .

3. There exists a constant $C > 0$ such that $\|L(v)\|_W \leq C \|v\|_V$ for all $v \in V$.

4. L is a Lipschitz continuous function on V .

Theorem (Theorem 5.5). Let $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ be normed spaces. Let $L(V, W)$ denote the space of all continuous linear mappings from V to W , equipped with the operator norm

$$\|T\|_{L(V, W)} := \sup_{\|v\|_V \leq 1} \|Tv\|_W.$$

If $(W, \|\cdot\|_W)$ is a Banach space, then the operator space $(L(V, W), \|\cdot\|_{L(V, W)})$ is also a Banach space (i.e. $L(V, W)$ is complete under the operator norm).

Hilbert Space Results

Corollary (Corollary 6.6). Let $(H, (\cdot, \cdot))$ be an inner product space. Then

$$\|u\| := \sqrt{(u, u)}$$

defines a norm on H , called the norm induced by the inner product.

Theorem (Theorem 6.10 (Parallelogram Identity)). Let $(V, \|\cdot\|)$ be a normed space. The norm $\|\cdot\|$ is induced by an inner product on V (i.e. V is a pre-Hilbert space) if and only if the following parallelogram identity holds for all $u, v \in V$:

$$\|u + v\|^2 + \|u - v\|^2 = 2\|u\|^2 + 2\|v\|^2.$$

In this case, one can define an inner product that induces $\|\cdot\|$ by the formula

$$(u, v) := \frac{1}{4} (\|u + v\|^2 - \|u - v\|^2)$$

in the real case (and a similar formula in the complex case).

Theorem (Theorem 6.14). Let $\{e_n\}_{n \in \mathbb{N}}$ be an orthonormal system in a separable Hilbert space $(H, (\cdot, \cdot))$. The following statements are equivalent:

1. $\{e_n\}_{n \in \mathbb{N}}$ is complete in H (i.e. an orthonormal basis of H).
2. $\|u\|^2 = \sum_{n=0}^{\infty} |(u, e_n)|^2$ for every $u \in H$ (Bessel's identity becomes an equality).
3. $(u, v) = \sum_{n=0}^{\infty} (u, e_n)(v, e_n)$ for all $u, v \in H$ (Parseval's identity).

Theorem (Theorem 6.16 (Gram-Schmidt Process)). Let $\{v_n\}_{n \geq 1}$ be a linearly independent sequence in an inner product space $(H, (\cdot, \cdot))$. Then there exists an orthonormal sequence $\{e_n\}_{n \geq 1}$ in H such that for each $N \in \mathbb{N}$, $\text{span}\{e_1, \dots, e_N\} = \text{span}\{v_1, \dots, v_N\}$. (That is, the orthonormal system $\{e_n\}$ is obtained from $\{v_n\}$ by the Gram-Schmidt procedure.)

Theorem (Theorem 6.17 (Riesz Representation)). Let H be a Hilbert space and H' its dual space. For every continuous linear functional $L \in H'$, there exists a unique vector $u_L \in H$ such that

$$L(v) = (u_L, v) \quad \forall v \in H,$$

and moreover $\|L\|_{H'} = \|u_L\|_H$. (In particular, H' is isometrically isomorphic to H itself.)

Theorem (Theorem 6.19 (Adjoint Operator)). Let $(V, (\cdot, \cdot)_V)$ and $(W, (\cdot, \cdot)_W)$ be two Hilbert spaces over the same field K . For each linear continuous operator $L \in L(V, W)$, there exists a unique linear continuous operator $L^* : W \rightarrow V$ (called the adjoint of L) such that

$$(L(u), w)_W = (u, L^*(w))_V \quad \forall u \in V, \forall w \in W.$$

3 Hahn–Banach Theorem and Its Consequences

Analytic Forms of Hahn–Banach

Theorem (Theorem 2.2 (Hahn–Banach, Analytic Form over $\mathbb{R}$)). *Let $(V, +, \cdot)$ be a real vector space. Let $p : V \rightarrow \mathbb{R}$ be a function satisfying*

$$p(\lambda v) = \lambda p(v) \quad \forall v \in V, \lambda > 0, \quad p(u + v) \leq p(u) + p(v) \quad \forall u, v \in V,$$

(i.e. p is a sublinear or quasi-norm). Let M be a linear subspace of V and let $\ell : M \rightarrow \mathbb{R}$ be a linear functional such that

$$\ell(v) \leq p(v) \quad \forall v \in M.$$

Then there exists an extension of ℓ to a linear functional $L : V \rightarrow \mathbb{R}$ (so $L|_M = \ell$) such that

$$L(v) \leq p(v) \quad \forall v \in V.$$

Theorem (Theorem 2.4 (Hahn–Banach, Analytic Form over $\mathbb{C}$)). *Let $(V, +, \cdot)$ be a complex vector space. Let $p : V \rightarrow \mathbb{R}$ be a sublinear function. Let M be a linear subspace of V and $\ell : M \rightarrow \mathbb{C}$ a linear functional such that*

$$\Re\{\ell(v)\} \leq p(v) \quad \forall v \in M.$$

Then there exists an extension of ℓ to a linear functional $L : V \rightarrow \mathbb{C}$ such that

$$\Re\{L(v)\} \leq p(v) \quad \forall v \in V.$$

Theorem (Theorem 3.1 (Hahn–Banach for Seminorms)). *Let $(V, +, \cdot)$ be a real or complex vector space. Let $p : V \rightarrow \mathbb{R}$ be a seminorm (i.e. p is sublinear and $p(\lambda v) = |\lambda|p(v)$ for all λ). Let M be a subspace of V and $\ell : M \rightarrow K$ a linear functional (over $K = \mathbb{R}$ or $\mathbb{C}$) such that*

$$|\ell(v)| \leq p(v) \quad \forall v \in M.$$

Then there exists a linear extension $L : V \rightarrow K$ of ℓ such that

$$|L(v)| \leq p(v) \quad \forall v \in V.$$

Hahn–Banach in Normed Spaces and Consequences

Theorem (Theorem 4.1 (Hahn–Banach in a Normed Space)). *Let $(V, \|\cdot\|_V)$ be a normed space over $K = \mathbb{R}$ or $\mathbb{C}$. Let M be a subspace of V and $\ell : M \rightarrow K$ a linear continuous functional (i.e. $\ell \in M'$). Then there exists a linear continuous functional $L : V \rightarrow K$ (i.e. $L \in V'$) such that*

$$L(v) = \ell(v) \quad \forall v \in M, \quad \text{and} \quad \|L\|_{V'} = \|\ell\|_{M'}.$$

Theorem (Theorem 4.3). *Let $V \neq \{0\}$ be a normed space. Then for every nonzero $v \in V$, there exists some $L_v \in V'$ such that*

$$L_v(v) = \|v\|_V, \quad \|L_v\|_{V'} = 1.$$

Consequently, V' (the dual space) is nonzero whenever V is nonzero.

Corollary (Corollary 4.4). *Let $V \neq \{0\}$ be a normed space. Then the norm of any vector can be computed as*

$$\|v\|_V = \sup_{\substack{L \in V' \\ L \neq 0}} \frac{|L(v)|}{\|L\|_{V'}} \quad \forall v \in V.$$

In particular, if $v \in V$ satisfies $L(v) = 0$ for every continuous linear functional $L \in V'$, then $v = 0$.

Theorem (Theorem 4.5). *Let $(V, \|\cdot\|_V)$ be a normed space over $K = \mathbb{R}$ or $\mathbb{C}$, and let M be a closed proper subspace of V (i.e. $M \subsetneq V$ is closed). Given any $v \in V \setminus M$, there exists a continuous linear functional $L_v \in V'$ such that*

$$L_v(u) = 0 \quad \forall u \in M, \quad L_v(v) = d(v, M) > 0, \quad \|L_v\|_{V'} = 1,$$

where $d(v, M) = \inf_{u \in M} \|v - u\|_V$ is the distance from v to M .

Corollary (Corollary 4.6). *Let $(V, \|\cdot\|_V)$ be a normed space and let M be a closed proper subspace of V . For any $v \in V \setminus M$, with $d := d(v, M) > 0$, there exists a continuous linear functional $T_v \in V'$ such that*

$$T_v(u) = 0 \quad \forall u \in M, \quad T_v(v) = 1, \quad \|T_v\|_{V'} = \frac{1}{d}.$$

Corollary (Corollary 4.7). *Let $(V, \|\cdot\|_V)$ be a normed space and let M be a subspace of V . If every continuous linear functional $L \in V'$ that vanishes on M is identically zero, then M is dense in V (equivalently $M = \overline{M} = V$).*

Geometric Forms of Hahn–Banach

Theorem (Theorem 5.4 (Hahn–Banach, First Geometric Form)). *Let $(V, \|\cdot\|_V)$ be a normed space over $K = \mathbb{R}$ or $\mathbb{C}$. Let E and F be two nonempty disjoint convex subsets of V , and assume that E is open. Then there exists a closed hyperplane $H = \{L = \alpha\}$ (for some $L \in V'$ and $\alpha \in K$) that separates E and F . In other words, there is $L \in V'$ and $\alpha \in \mathbb{R}$ such that*

$$L(v) \leq \alpha \quad \forall v \in E, \quad L(w) \geq \alpha \quad \forall w \in F.$$

Lemma (Lemma 5.5). *Let $(V, \|\cdot\|_V)$ be a normed space over $K = \mathbb{R}$ or $\mathbb{C}$. Let $C \subset V$ be a convex open set containing 0_V . Define the Minkowski functional of C by*

$$\rho_C(v) := \inf\{\lambda > 0 : \lambda^{-1}v \in C\}.$$

Then $\rho_C : V \rightarrow [0, \infty)$ is a quasi-norm (sublinear and positively homogeneous), and moreover one has

$$C = \{v \in V : \rho_C(v) < 1\}.$$

Lemma (Lemma 5.6). *Let $(V, \|\cdot\|_V)$ be a normed space and let $C \subset V$ be a nonempty open convex proper subset. If $v_0 \in V \setminus C$, then there exists a continuous linear functional $L \in V'$ such that*

$$L(v) < L(v_0) \quad \forall v \in C.$$

In particular, the hyperplane $H = \{v \in V : L(v) = L(v_0)\}$ separates the point $\{v_0\}$ and the set C .

Theorem (Theorem 5.7 (Hahn–Banach, Second Geometric Form)). *Let $(V, \|\cdot\|_V)$ be a normed space over $K = \mathbb{R}$ or $\mathbb{C}$. Let E and F be two nonempty disjoint convex subsets of V , with E closed and F compact. Then there exists a closed hyperplane $H = \{L = \alpha\}$ (for some $L \in V'$ and $\alpha \in \mathbb{R}$) that strictly separates E and F . That is, there exists $\varepsilon > 0$ and $L \in V'$ such that*

$$L(v) \leq \alpha - \varepsilon \quad \forall v \in E, \quad L(w) \geq \alpha + \varepsilon \quad \forall w \in F.$$

Corollary (Corollary 5.8). *Let $(V, \|\cdot\|_V)$ be a normed space and let $A \subset V$ be a proper subspace ($A \neq V$). Then there exists a nonzero continuous linear functional $L \in V'$ such that $L(v) = 0$ for all $v \in A$.*

4 Baire Category and Banach's Theorems

Baire's Theorem

Theorem (Theorem 1.1 (Baire Category Theorem)). *Let (X, d) be a complete metric space. If $(O_n)_{n \in \mathbb{N}}$ is a sequence of dense open subsets of X , then*

$$\bigcap_{n \in \mathbb{N}} O_n$$

is dense in X (and in particular is nonempty).

Corollary (Corollary 1.3). *Let (X, d_X) be a complete metric space which can be written as a countable union of closed sets: $X = \bigcup_{n \in \mathbb{N}} C_n$. Then*

$$\Omega := \bigcup_{n \in \mathbb{N}} \text{Int}(C_n)$$

is a dense open subset of X (in fact $\Omega = X$). Equivalently, at least one of the closed sets C_n must have a nonempty interior.

Uniform Boundedness Principle

Theorem (Theorem 3.1 (Banach–Steinhaus Theorem)). *Let $(V, \|\cdot\|_V)$ be a Banach space and $(W, \|\cdot\|_W)$ a normed space over the same field. Let $\{T_i : V \rightarrow W\}_{i \in I}$ be a family of continuous linear operators such that for each $v \in V$, the set $\{\|T_i(v)\|_W : i \in I\}$ is bounded. Then the operator norms of the T_i are uniformly bounded; in fact*

$$\sup_{i \in I} \|T_i\|_{L(V, W)} < \infty.$$

Corollary (Corollary 3.2). *Let $(V, \|\cdot\|_V)$ be a Banach space and $(W, \|\cdot\|_W)$ a normed space, and let $T_n : V \rightarrow W$ be a sequence of continuous linear operators. Assume that for each $v \in V$, the sequence $T_n(v)$ converges in W to some limit denoted $T(v)$. Then the limit map $T : V \rightarrow W$ defined by $T(v) = \lim_{n \rightarrow \infty} T_n(v)$ is a continuous linear operator in $L(V, W)$. (In particular, pointwise limits of bounded linear maps are bounded linear.)*

Open Mapping Theorem

Theorem (Theorem 4.3 (Banach Open Mapping Theorem)). *Let $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ be two Banach spaces over the same field K . If $T : V \rightarrow W$ is a surjective linear continuous operator, then T is an open map. In other words, there exists $\delta > 0$ such that*

$$T(B_V(0, 1)) \supset B_W(0, \delta),$$

where $B_V(0, 1)$ and $B_W(0, \delta)$ are the balls of radius 1 and δ around 0 in V and W , respectively.

Corollary (Corollary 4.4). *Let $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ be Banach spaces, and let $T : V \rightarrow W$ be a bijective linear continuous map. Then $T^{-1} : W \rightarrow V$ is also continuous. (In particular, any bijective bounded linear operator between Banach spaces is an isomorphism of Banach spaces.)*

Closed Graph and Hellinger–Toeplitz Theorems

Theorem (Theorem 5.1 (Banach Closed Graph Theorem)). *Let $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ be Banach spaces. Suppose $T : V \rightarrow W$ is a linear mapping such that the graph*

$$G_T := \{(v, w) \in V \times W : w = Tv\}$$

is a closed subset of the product space $V \times W$ (with the norm $N((v, w)) := \|v\|_V + \|w\|_W$). Then T is continuous (bounded). Equivalently: if T is a linear map for which $v_n \rightarrow v$ in V and $Tv_n \rightarrow w$ in W together imply $w = Tv$, then T must be a bounded operator.

Theorem (Theorem 5.2 (Hellinger–Toeplitz Theorem)). *Let $(H, (\cdot, \cdot))$ be a Hilbert space, and let $T : H \rightarrow H$ be a linear operator which is symmetric, meaning $(Tv, w) = (v, Tw)$ for all $v, w \in H$. Then T is continuous (i.e. bounded as an operator on H).*

5 Projection on Convex Sets and Variational Inequalities

Projection onto a Convex Set

Theorem (Theorem 2.1 (Projection onto a Closed Convex Subset)). *Let $(H, (\cdot, \cdot))$ be a Hilbert space over $K = \mathbb{R}$ or $\mathbb{C}$, with induced norm $\|x\|_H = \sqrt{(x, x)}$. Let $K \subset H$ be a nonempty closed convex subset, and fix $u \in H$. Then there exists a unique $u_K \in K$ such that*

$$\|u - u_K\|_H = \inf_{v \in K} \|u - v\|_H.$$

Moreover, this minimizing vector u_K is characterized by the orthogonality condition

$$(u - u_K, v - u_K) \leq 0 \quad \forall v \in K.$$

In particular, defining $P_K : H \rightarrow H$ by $P_K(u) := u_K$, we obtain a mapping P_K such that:

[(a)]

1. $P_K(H) = K$ (range is K);
2. $\|P_K(u_1) - P_K(u_2)\|_H \leq \|u_1 - u_2\|_H$ for all $u_1, u_2 \in H$ (P_K is 1-Lipschitz, in fact nonexpansive).
3. P_K is idempotent: $P_K(u) = u$ for all $u \in K$, and hence $P_K(P_K(u)) = P_K(u)$ for all $u \in H$.

Stampacchia's Theorem (Variational Inequality)

Theorem (Theorem 2.3 (Stampacchia's Existence Theorem)). *Let $(H, (\cdot, \cdot))$ be a real Hilbert space with dual H' and induced norm $\|\cdot\|_H$. Let $K \subset H$ be a nonempty closed convex subset. Let $\ell \in H'$ be a continuous linear functional, and let $a(\cdot, \cdot) : H \times H \rightarrow \mathbb{R}$ be a continuous bilinear form which is coercive and symmetric on H . Then there exists a unique $u \in K$ such that*

$$a(u, v - u) \geq \langle \ell, v - u \rangle \quad \forall v \in K,$$

i.e. u solves the variational inequality for a and ℓ on the constraint set K .

Lax–Milgram Theorem

Theorem (Theorem 2.6 (Lax–Milgram Theorem)). *Let $(H, (\cdot, \cdot))$ be a real Hilbert space. Let $a(\cdot, \cdot) : H \times H \rightarrow \mathbb{R}$ be a continuous bilinear form which is coercive (not necessarily symmetric). Let $\ell \in H'$ be a continuous linear functional on H . Then there exists a unique $u \in H$ such that*

$$a(u, v) = \langle \ell, v \rangle \quad \forall v \in H.$$

In other words, the linear operator $B : H \rightarrow H'$ defined by $v \mapsto [w \mapsto a(v, w)]$ is bijective, so for each $\ell \in H'$ there is a unique solution $u \in H$.